

Supplement: empirical probes and cross-disciplinary reframings of the Goldbach–Chen frontier

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This supplement records, in compact form, the cross-disciplinary reframings of the main paper, *The Goldbach–Chen Frontier: A Singular-Series Cancellation behind a Finite-Range Half-Power*, together with the empirical probes moved out of the body. Each reframing recasts the *same* singular-series governance in a different external language; we keep them for the individual vocabularies and the one honest (structural) negative, not as independent results. “The main text” refers to that paper, and every number is reproducible from the `run_*.py` drivers in the repository.

1 An honest structural negative

The frontier admits many external recastings—thermodynamic (an arithmetic ground state), network (adversarial robustness), control (value of information), information-theoretic, and geometric. Each is the same fact— $\mathfrak{S}(N)$ sets the local densities—seen in a different mirror, with no mathematical content of its own, so we do not develop them (they are reproducible from the `run_*.py` drivers). One recasting, however, returns a genuine *structural* negative worth recording, because it bounds what the additive structure does *not* buy. *Persistent topology*: for the binary problem the additive-complexity sublevel filtration is a disjoint matching, so its 0-dimensional persistence is just the energy spectrum and all higher homology vanishes—genuine higher-order topological structure appears only at the ternary level, which is anti-clustered.

2 Further empirical probes (moved from the main text)

For length and focus, the following experiments were moved out of the main text. Each is reproducible from its `run_*.py` driver in the repository and, like every reframing here, returns to the same singular series $\mathfrak{S}(N)$.

2.1 The Chen-rescue threshold

Restricting representations to a balanced band $p \in [N/2 - k, N/2]$ and measuring the fraction of N that *require* a semiprime (no prime pair in the band) gives a unimodal curve in k (Fig. 1). For $k \gtrsim 0.005N$ the prime channel essentially never fails; the rescue fraction peaks near 38–40% at $k \sim 20\text{--}40$; for $k = 1, 2$ it falls again because often no representation exists at all. The peak shifts right with N (roughly as $\log^2 N$), where the expected number of balanced prime pairs $\sim 2C_2 \mathfrak{S}(N) k / \log^2(N/2)$ drops to $O(1)$. This is the regime captured by the residual exponent $a_{\text{Res}}(N)$ of the main text: Chen’s flexibility becomes *necessary* precisely when near-perfect balance is imposed, so the rescue threshold and a_{Res} track the same singular series.

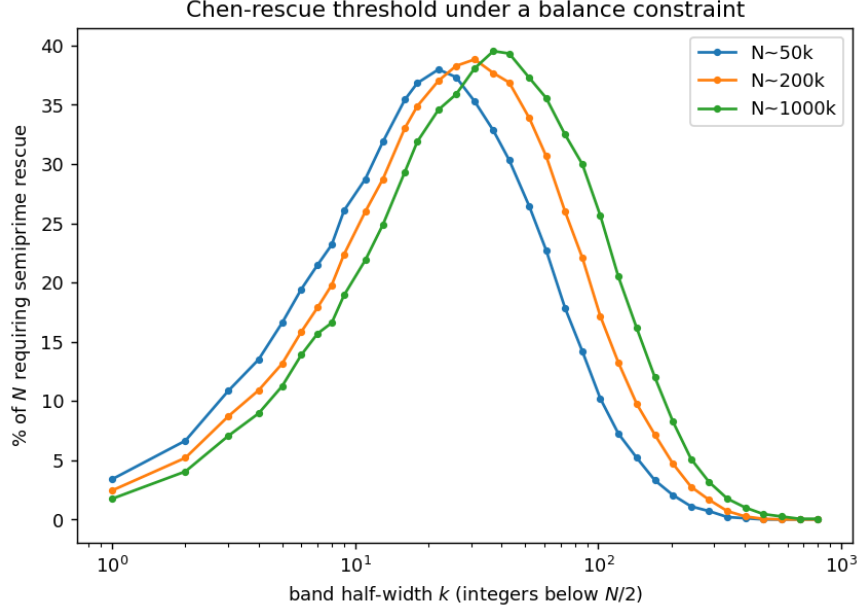


Figure 1: Fraction of N that require a semiprime rescue versus band half-width k , for three scales of N . The curve is unimodal with peak scaling like $\log^2 N$.

2.2 Weak continuity of the location measures

The pointwise measure μ_N of q/N is noisy, but averaging over a window $N \in [X, X + H]$ produces a stable density (Fig. 2). At $X = 10^6$: (i) splitting a window into two same-scale subsamples, the total-variation distance between their densities falls monotonically from $2.1 \cdot 10^{-3}$ to $3 \cdot 10^{-4}$ as H grows from 125 to 4000 (a concrete form of weak convergence); (ii) the prime- q density is within total variation 0.028 of uniform on $(0, 1)$, with a mild endpoint upturn (the $1/(\log(xN)\log((1-x)N))$ effect); (iii) prime and semiprime locations agree to total variation 0.019, the semiprime location slightly skewed toward larger q (mean $q/N = 0.513$ versus 0.500 for primes). This probe earns its place because it is explained by the deferred analytic model of §3.1 below: under the weighted Hardy–Littlewood asymptotics, after normalization both the inherited singular series *and* the Mertens channel sum cancel, leaving the uniform law for q/N in either channel—the same cancellation that drives the whole paper, now visible in the location law.

2.3 Topology of valleys

Normalising each count by its own singular factor ($R_1/\mathfrak{S}$, $R_{\leq 2}/(\mathfrak{S} + \mathfrak{S}^{1/2}W)$) and computing the 0-dimensional sublevel persistence of the detrended sequences, the valleys of the Goldbach and Chen comets largely *coincide*: the anomalies correlate at +0.64, and at the 200 most fragile N the Chen anomaly (0.29) is as deep as the Goldbach one (0.30). Chen does *not* topologically fill the Goldbach valleys (only the deepest 30% are softened, Fig. 3); the rescue is one of magnitude, not of valley structure, because both channels share the constraint that p be prime. The correlation +0.64 is a within-window descriptive statistic over a strongly arithmetically dependent sequence (consecutive even N), not an i.i.d. estimate.

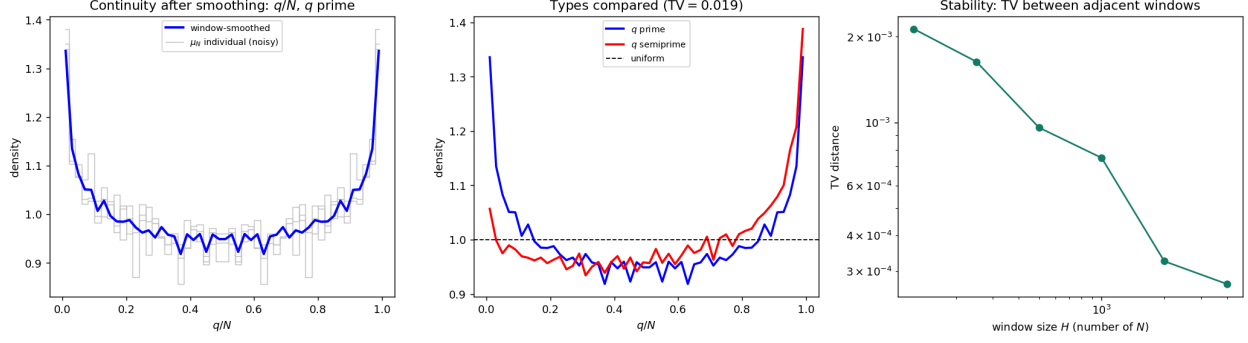


Figure 2: Weak continuity of q/N . Left: noisy pointwise μ_N versus the window-smoothed density. Center: prime versus semiprime location densities, both near uniform. Right: same-scale sampling distance versus window size H (log–log).

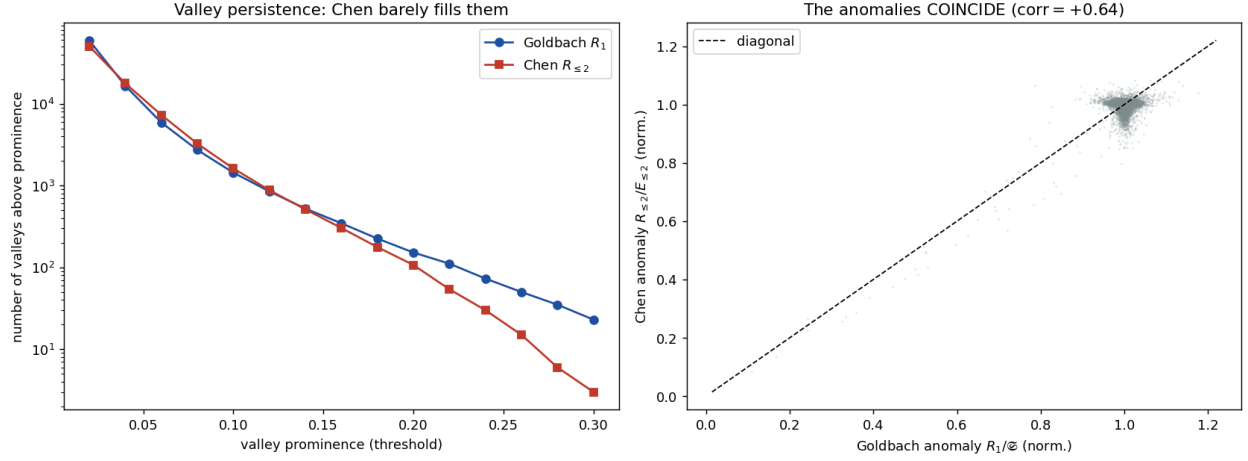


Figure 3: Valley persistence. Left: Chen has nearly as many valleys as Goldbach. Right: the normalised anomalies coincide (+0.64).

2.4 Statistical dynamics of the frontier

The prime share is almost a deterministic function of the singular series: $\theta(N)$ is 94% explained by $\mathfrak{S}(N)$ alone (adding $N \bmod 30$ contributes nothing further), and the increment $\Delta\theta_N$ is 98% explained by the jump $\mathfrak{S}(N) \rightarrow \mathfrak{S}(N+2)$. The residual is small (25% of the spread of θ) and is *not* white noise (lag-1 autocorrelation -0.21 after detrending): the frontier carries little irreducible randomness, being governed by the local arithmetic of N (Fig. 4). These percentages are descriptive R^2 values over an arithmetically dependent window, not i.i.d. fits.

3 Deferred analytic models (from the main appendix)

These two model/conditional results were moved here from the main appendix to keep it to three parts; they explain, respectively, the weak-continuity probe and the carry signature of the main text.

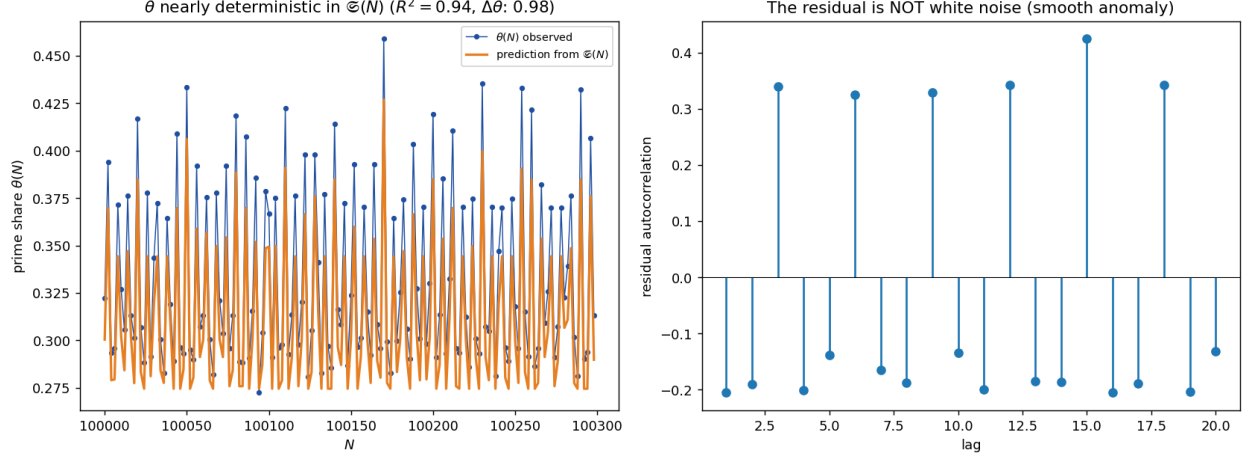


Figure 4: Dynamics. Left: $\theta(N)$ tracks its singular-series prediction. Right: the residual autocorrelation, small and non-white.

3.1 Weak uniformity of the location measures

The continuity experiments (supplement) suggest that the representation clouds become continuous after normalization: the empirical laws of q/N for both prime and semiprime q approach the uniform law on $(0, 1)$, up to small endpoint effects. This is what the weighted Hardy–Littlewood heuristic predicts. For a bounded C^1 test function $\varphi : [0, 1] \rightarrow \mathbb{R}$ write, when $R_t(N) \neq 0$,

$$\mu_N^{(1)}(\varphi) := \frac{1}{R_1(N)} \sum_{\substack{p < N \\ p, N-p \in \mathbb{P}}} \varphi\left(\frac{N-p}{N}\right), \quad \mu_N^{(2)}(\varphi) := \frac{1}{R_2(N)} \sum_{\substack{p < N \\ p \in \mathbb{P}, N-p \in \mathcal{S}_2}} \varphi\left(\frac{N-p}{N}\right),$$

the averages of φ against the type- t location measures.

Hypothesis 1 (Weighted prime and semiprime channel asymptotics). *For every bounded C^1 test function φ and almost all even N ,*

$$\sum_{\substack{p < N \\ p, N-p \in \mathbb{P}}} \varphi\left(\frac{N-p}{N}\right) = (1 + o(1)) 2C_2 \mathfrak{S}(N) \frac{N}{\log^2 N} \int_0^1 \varphi(x) dx,$$

and, for each unblocked semiprime channel $q = rs$, uniformly in the range of r used in the paper,

$$\sum_{\substack{s \in \mathbb{P} \\ N-rs \in \mathbb{P}}} \varphi\left(\frac{rs}{N}\right) = (1 + o(1)) 2C_2 \mathfrak{S}(N) \frac{r-1}{r-2} \frac{N}{r \log^2 N} \int_0^1 \varphi(x) dx,$$

the channel omitted when $r \mid N$.

Proposition 1 (Weak uniformity of the location measures; conditional). *Under Hypothesis 1, for almost all even N ,*

$$\mu_N^{(1)}(\varphi) = \int_0^1 \varphi(x) dx + o(1), \quad \mu_N^{(2)}(\varphi) = \int_0^1 \varphi(x) dx + o(1).$$

Equivalently, the empirical measures of q/N in the prime and semiprime channels converge weakly to the uniform law on $(0, 1)$, in the same smoothed/almost-all sense as the weighted asymptotics.

Proof. For the prime channel, the first weighted asymptotic with $\varphi \equiv 1$ gives $R_1(N) = (1 + o(1)) 2C_2 \mathfrak{S}(N) N / \log^2 N$; dividing the two estimates yields $\mu_N^{(1)}(\varphi) = \int_0^1 \varphi + o(1)$. For the semiprime channel, writing $q = rs$ and summing the per-channel asymptotic over the unblocked $r \leq \sqrt{N}$,

$$\sum_{\substack{p < N \\ p \in \mathbb{P}, N-p \in \mathcal{S}_2}} \varphi\left(\frac{N-p}{N}\right) = (1 + o(1)) 2C_2 \mathfrak{S}(N) \frac{N}{\log^2 N} \left(\sum_{\substack{3 \leq r \leq \sqrt{N} \\ r \nmid N}} \frac{r-1}{r} \frac{1}{r} \right) \int_0^1 \varphi(x) dx,$$

the parenthesis being the Mertens channel sum $W_f(N)$. Taking $\varphi \equiv 1$ gives the matching asymptotic for $R_2(N)$; dividing cancels both the inherited singular series $\mathfrak{S}(N)$ and the channel sum, leaving $\mu_N^{(2)}(\varphi) = \int_0^1 \varphi + o(1)$. This is weak convergence against every bounded C^1 test function. $\square$

This explains why the discontinuous pointwise representation sets nonetheless produce smooth limiting clouds: the singular series governs the local *density* of representations, but after normalizing each measure it cancels—together with the Mertens channel sum in the semiprime case—leaving the uniform spatial law. It is the conditional counterpart of the weak-continuity diagnostics (supplement), and like the other appendix results it is conditional on the stated weighted asymptotics, not an unconditional theorem about primes.

3.2 A last-digit model for the carry excess

The carry geometry of the carry observation in the main text can be partly explained by a local digit calculation: prime-like summands are not uniform among terminal digits but restricted to reduced residue classes, and this restriction alone creates a positive carry bias in the least significant digit. Fix a base $b \geq 2$ and let $U_b := \{1 \leq u \leq b-1 : (u, b) = 1\}$ be the reduced residue digits modulo b . For digits $x, y \in \{0, \dots, b-1\}$ let $\kappa_b(x, y) := \mathbf{1}_{\{x+y \geq b\}}$ be the carry out of the least significant digit.

Proposition 2 (Local carry excess; model). *Let X, Y be independent uniform digits in $\{0, \dots, b-1\}$ and U, V independent uniform digits in U_b . Then*

$$\mathbb{E} \kappa_b(U, V) = \frac{1}{2} + \frac{1}{2} \Pr(U + V = b), \quad \mathbb{E} \kappa_b(X, Y) = \frac{b-1}{2b},$$

so $\mathbb{E} \kappa_b(U, V) - \mathbb{E} \kappa_b(X, Y) = \frac{1}{2b} + \frac{1}{2} \Pr(U + V = b) > 0$: reduced-residue summands have a strictly larger least-digit carry probability than unrestricted ones.

Proof. For unrestricted digits, $\#\{(x, y) : 0 \leq x, y \leq b-1, x+y \geq b\} = \sum_{x=1}^{b-1} x = \frac{b(b-1)}{2}$, so $\mathbb{E} \kappa_b(X, Y) = (b-1)/(2b)$. For reduced residues, the involution $(u, v) \mapsto (b-u, b-v)$ maps U_b^2 to itself (since $(u, b) = 1 \iff (b-u, b) = 1$), sends $\{u+v < b\}$ bijectively to $\{u+v > b\}$, and fixes $\{u+v = b\}$; hence those two strict sets are equinumerous and $\Pr(U + V \geq b) = \frac{1}{2} + \frac{1}{2} \Pr(U + V = b)$. $\square$

The calculation is deliberately local: it does not prove the full carry statistics of actual Goldbach pairs, nor assume independence of higher digits. It shows that the most basic prime residue restriction already pushes the carry count up. In base 2 both odd prime summands end in 1, so the least digit always carries; in larger bases the effect survives after averaging over reduced classes. Since the total carry count obeys the Kummer–Legendre identity $\#\text{carries}_b(a, q) = (S_b(a) + S_b(q) - S_b(N))/(b-1)$ (the carry observation in the main text), the proposition is a positive *local* contribution to it, and the empirical excess of the carry observation in the main text reads as its multi-digit continuation.

Remark 1 (Fixed- N caveat). *For a fixed terminal digit $N \bmod b$ the carry bias can depend on the residue class and need not be positive in every class; the proposition is an averaged local-prime model, explaining why a positive excess is natural on average while the full statistic remains a global multi-digit measurement. Like the other appendix results it is model-based, not a theorem about the empirical carry law.*